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1. Introduction

In today's digital transportation industry, companies like **OLA** generate a large amount of ride-booking data every day. This data includes information about bookings, customers, drivers, vehicle types, payments, cancellations, and ride ratings. However, raw data alone does not provide value unless it is properly analyzed and converted into meaningful insights.

The **OLA Ride Data Analysis Project** focuses on analyzing ride-booking data using **SQL** and **Power BI** to understand overall business performance and customer behavior. The main goal of this project is to identify booking trends, analyze ride success and cancellations, evaluate customer and driver ratings, study payment method usage, and measure revenue performance.

In this project, **SQL** is used to extract and analyze data by answering real business questions such as:

- How many rides are successfully completed?
- Why do rides get cancelled or remain incomplete?
- Which customers and vehicle types contribute the most to revenue?

- How satisfied are customers and drivers based on ratings?

The analyzed results are then presented through an **interactive Power BI dashboard**, which makes the insights easy to understand for business stakeholders. This dashboard helps decision-makers quickly identify patterns, problems, and opportunities for improvement.

Overall, this project demonstrates how data analysis can help improve operational efficiency, enhance customer experience, and support data-driven decision-making in a ride-hailing business.

2. Problem Statement

OLA handles a large number of ride bookings every day across different vehicle types, customers, and payment methods. Due to the high volume of data, it becomes difficult for the business to clearly understand booking performance, cancellation reasons, customer behavior, driver performance, and overall revenue trends by simply looking at raw data.

Without proper analysis, key business questions remain unanswered, such as:

- How many rides are successfully completed versus cancelled or incomplete?
- What are the main reasons for ride cancellations?
- Which customers and vehicle types contribute the most to revenue?
- How satisfied are customers and drivers based on ratings?
- Which payment methods are most commonly used?

The lack of structured analysis and visual reporting makes it challenging for stakeholders to identify operational issues, improve customer experience, and make data-driven decisions.

This project addresses these challenges by using **SQL** to analyze raw ride-booking data and **Power BI** to present the results through an interactive dashboard. The solution helps transform complex data into clear insights that support better decision-making, operational improvements, and business growth.

3. Project Objectives

The main objective of this project is to analyze OLA ride-booking data and convert raw data into meaningful business insights using **SQL** and **Power BI**. The specific objectives of this project are:

- To analyze the overall booking performance, including successful, cancelled, and incomplete rides.
- To identify key reasons for ride cancellations and understand operational issues.
- To study customer behavior by analyzing booking patterns and top customers.
- To evaluate vehicle type performance based on ride distance and customer ratings.
- To analyze customer and driver ratings to measure service quality and satisfaction.
- To examine payment method usage and identify popular digital payment options.
- To calculate total booking value and understand revenue trends.
- To create an interactive Power BI dashboard that visually presents insights for easy decision-making.

4. Dataset Overview

The dataset contains ride-level booking information including booking details, customer and driver ratings, payment methods, ride distance, booking value, and cancellation reasons.

5. Tools & Technologies

- SQL
- Power BI
- MS Excel
- GitHub

6. SQL Analysis & Queries

```
1 • create database ola;  
2 • use ola;  
3  
4 • SELECT * FROM bookings WHERE Booking_Status = 'Success';  
5
```

Output:-

Result Grid										
Filter Rows: Export: Wrap Cell Content: Fetch rows:										
Date	Time	Booking_ID	Booking_Status	Customer_ID	Vehicle_Type	Pickup_Location	Drop_Location	V_TAT	C_TAT	Canceled_Rides
	22:20:00	CNR2940424040	Success	CID225428	Bike	Magadi Road	Varthur	203	30	NULL
2024-07-30 19:59:00	19:59:00	CNR2982357879	Success	CID270156	Prime SUV	Sahakar Nagar	Varthur	238	130	NULL
2024-07-02 9:02:00	9:02:00		Success	CID939555	Mini	Rajajinagar	Chamarajpet	252	80	NULL
2024-07-13 4:42:00	4:42:00	CNR8787177882	Success	CID802429	Mini	Kadugodi	Vijayanagar	231	90	NULL
2024-07-23 9:51:00	9:51:00	CNR3612067560	Success	CID476071	Bike	Tumkur Road	Whitefield	133	40	NULL
2024-07-29 23:33:00	23:33:00	CNR4787583516	Success	CID923404	Prime Plus	Hosur Road	Jayanagar	35	55	NULL
2024-07-26 4:03:00	4:03:00	CNR7943634301	Success	CID647026	Prime Plus	Kammanahalli	Rajajinagar	238	95	NULL
2024-07-27 13:18:00	13:18:00	CNR4524472111	Success	CID540929	Auto	Cox Town	Yelahanka	126	35	NULL
2024-07-16 9:54:00	9:54:00	CNR8181602032	Success	CID167642	Bike	Indiranagar	MG Road	70	95	NULL
2024-07-02 10:25:00	10:25:00	CNR8090918544	Success	CID640151	Bike	Magadi Road	HSR Layout	126	95	NULL
2024-07-05 23:47:00	23:47:00	CNR3106156650	Success	CID743775	Bike	Electronic City	Landford Town	140	40	NULL

Filter objects

bookings

Columns

Date
Time
Booking_ID
Booking_Status
Customer_ID
Vehicle_Type

1 • create database ola;
2 • use ola;
3
4 • SELECT Booking_ID, Incomplete_Rides_Reason FROM bookings WHERE Incomplete_Rides = 'Yes';
5

Result Grid

Filter Rows:

Export

	Booking_ID	Incomplete_Rides_Reason
▶	CNR5176704322	Customer Demand
	CNR9312632867	Vehicle Breakdown
	CNR7924302885	Customer Demand
	CNR1640228587	Other Issue
	CNR7623690602	Other Issue
	CNR9590311980	Customer Demand
	CNR5863244684	Customer Demand
	CNR9526078867	Customer Demand
	CNR7154043084	Customer Demand
	CNR3193710797	Other Issue
	CNR7073850950	Customer Demand
	CNR9952584604	Customer Demand

```

1 • create database ola;
2 • use ola;
3
4 • SELECT Customer_ID, COUNT(Booking_ID) as total_rides FROM bookings GROUP BY
5 Customer_ID ORDER BY total_rides DESC LIMIT 5;

```

Result Grid		
	Filter Rows:	
	Export:	Wrap Cell Content: Fetch rows:
Customer_ID	total_rides	
CID954071	5	
CID539191	4	
CID189965	4	
CID268274	4	
CID952434	4	

1. Retrieve all successful bookings:

- `SELECT * FROM bookings WHERE Booking_Status = 'Success';`

2. Find the average ride distance for each vehicle type:

- `SELECT Vehicle_Type, AVG(Ride_Distance) as avg_distance FROM bookings GROUP BY Vehicle_Type;`

3. Get the total number of cancelled rides by customers:

- `SELECT COUNT(*) FROM bookings WHERE Booking_Status = 'cancelled by Customer';`

4. List the top 5 customers who booked the highest number of rides:

- `SELECT Customer_ID, COUNT(Booking_ID) as total_rides FROM bookings GROUP BY Customer_ID ORDER BY total_rides DESC LIMIT 5;`

5. Get the number of rides cancelled by drivers due to personal and car-related issues:

- `SELECT COUNT(*) FROM bookings WHERE cancelled_Rides_by_Driver = 'Personal & Car related issue';`

6. Find the maximum and minimum driver ratings for Prime Sedan bookings:

- `SELECT MAX(Driver_Ratings) as max_rating, MIN(Driver_Ratings) as min_rating FROM bookings WHERE Vehicle_Type = 'Prime Sedan';`

7. Retrieve all rides where payment was made using UPI:

- `SELECT * FROM bookings WHERE Payment_Method = 'UPI';`

8. Find the average customer rating per vehicle type:

- `SELECT Vehicle_Type, AVG(Customer_Rating) as avg_customer_rating FROM bookings GROUP BY Vehicle_Type;`

9. Calculate the total booking value of rides completed successfully:

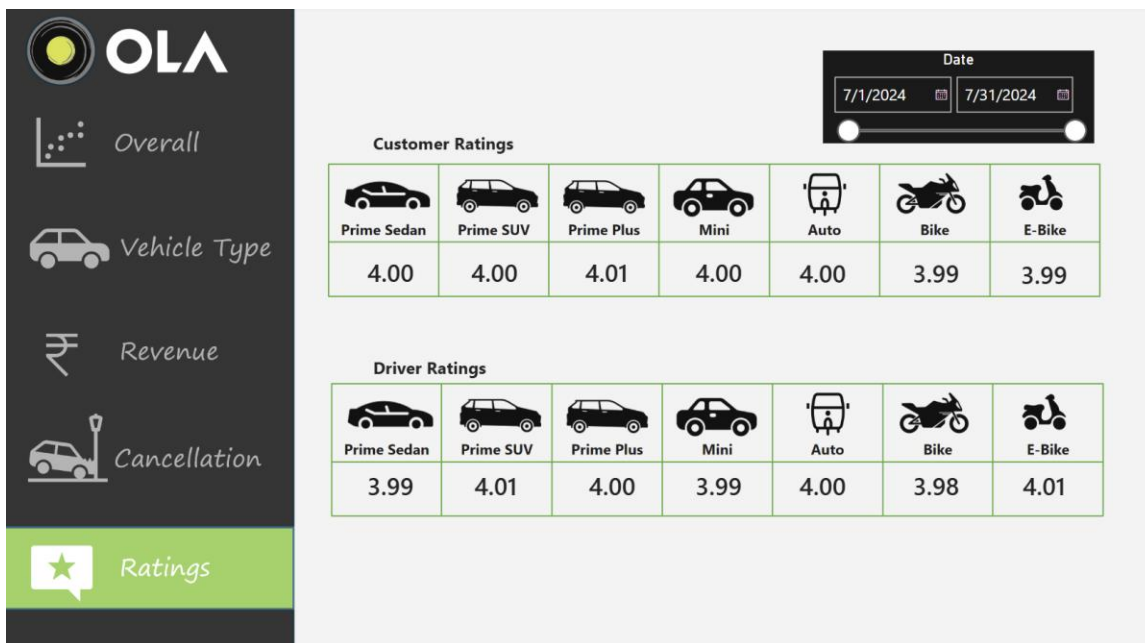
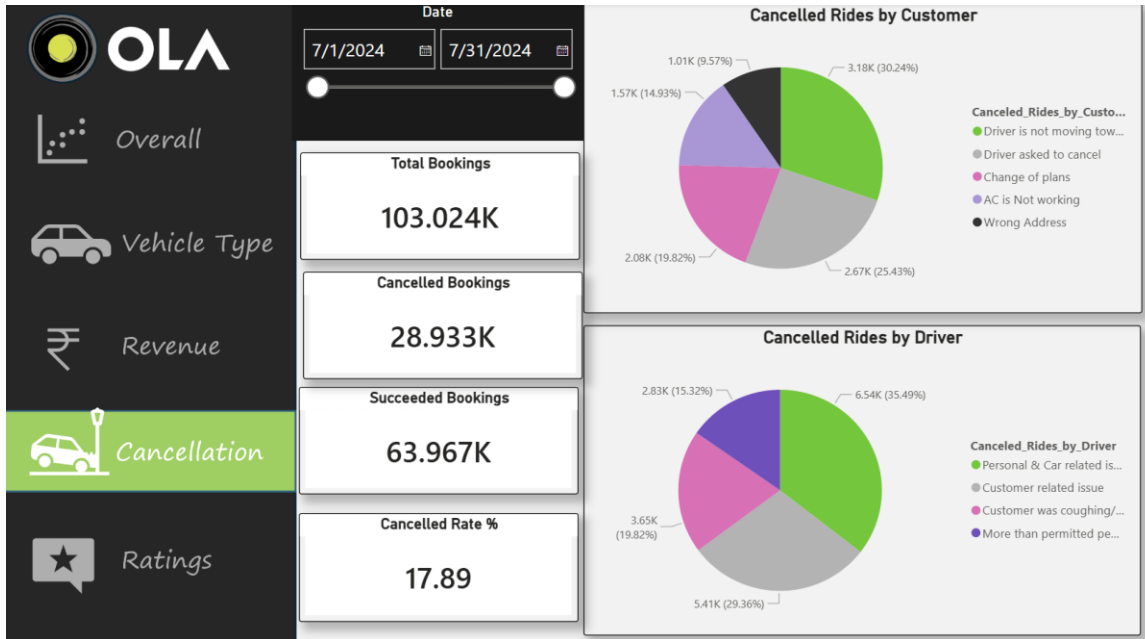
- `SELECT SUM(Booking_Value) as total_successful_value FROM bookings WHERE Booking_Status = 'Success';`

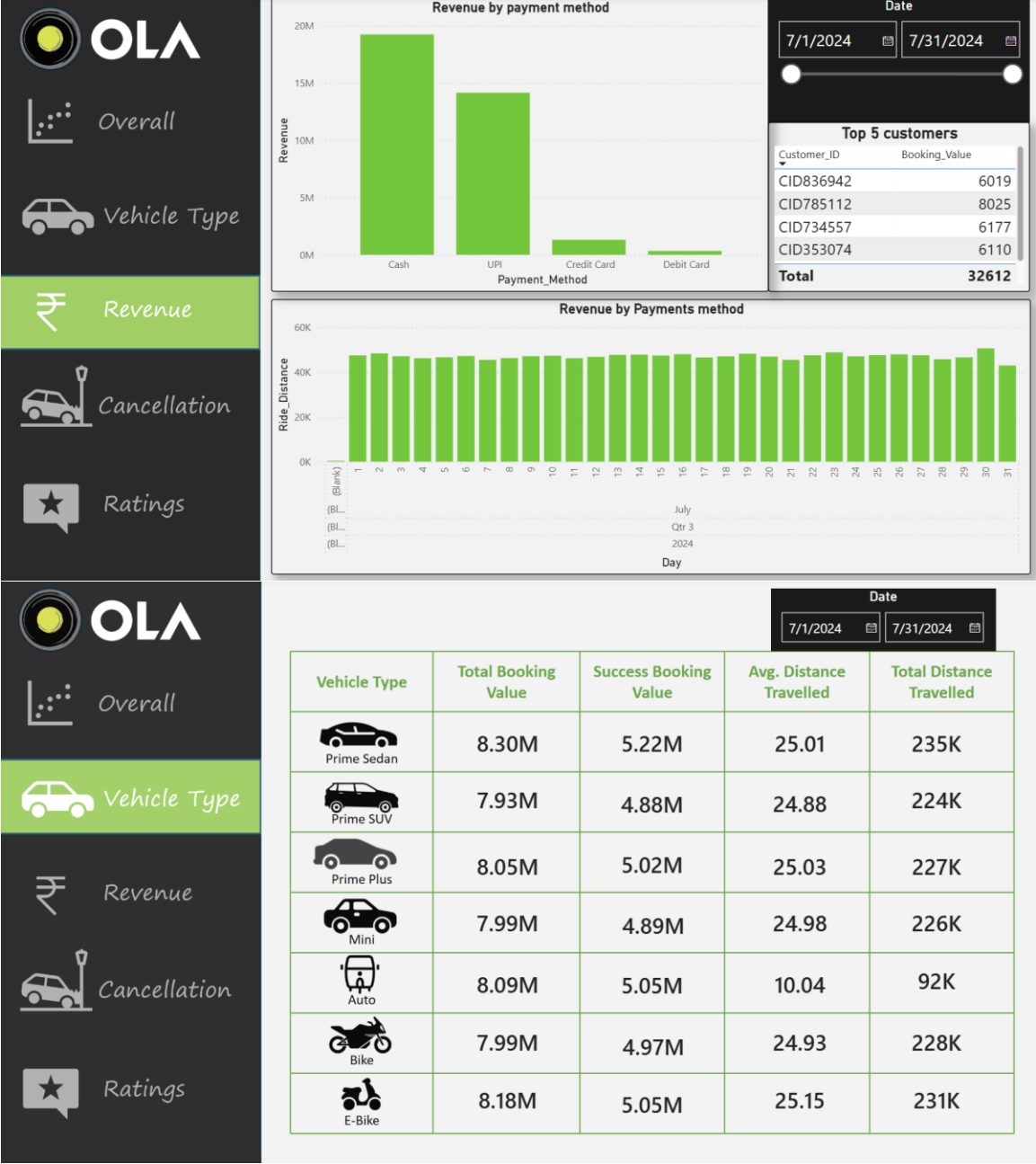
10. List all incomplete rides along with the reason:

- `SELECT Booking_ID, Incomplete_Rides_Reason FROM bookings WHERE Incomplete_Rides = 'Yes';`

7. Power BI Dashboard

An interactive Power BI dashboard was created to visualize booking trends, customer behavior, revenue, and ratings. The dashboard helps stakeholders quickly understand performance and identify issues.

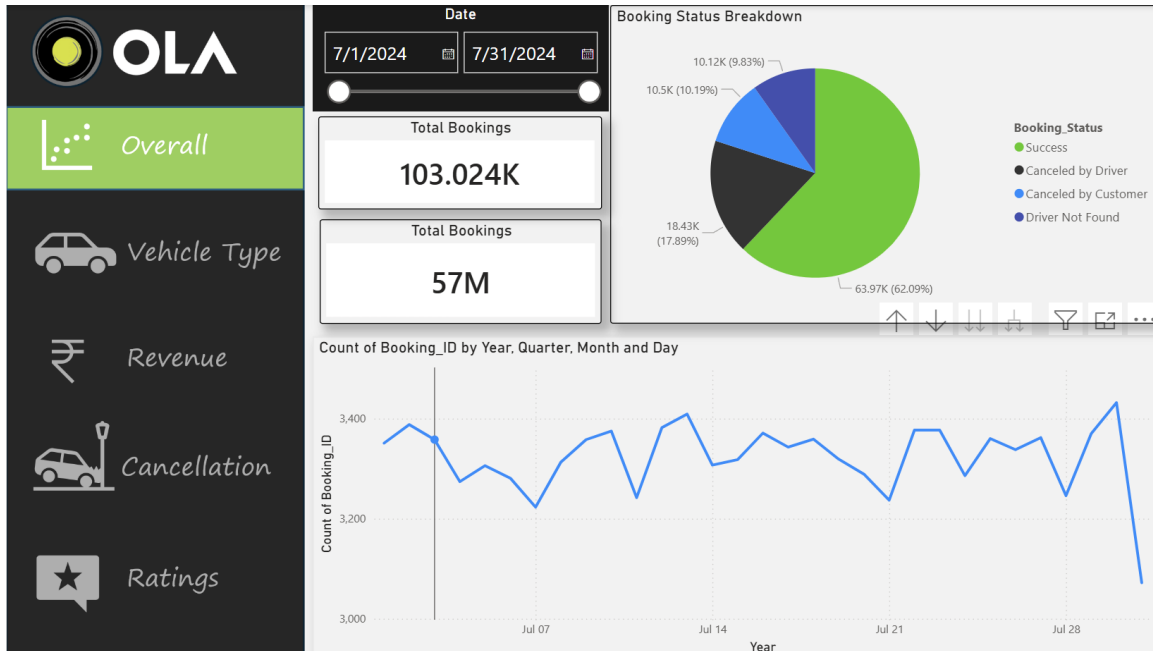




7/1/2024

7/31/2024

Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	8.30M	5.22M	25.01	235K
Prime SUV	7.93M	4.88M	24.88	224K
Prime Plus	8.05M	5.02M	25.03	227K
Mini	7.99M	4.89M	24.98	226K
Auto	8.09M	5.05M	10.04	92K
Bike	7.99M	4.97M	24.93	228K
E-Bike	8.18M	5.05M	25.15	231K



- Ride Volume Over Time – identifies peak and low-demand periods
- Booking Status Breakdown – shows success vs cancellations
- Revenue by Payment Method – highlights digital payment adoption
- Top Customers – identifies high-value users
- Ratings Analysis – compares customer and driver satisfaction

8. Key Insights & Outcomes

- A small group of customers contributes a significant portion of revenue
- UPI is the most preferred payment method
- Certain vehicle types receive higher customer ratings
- Driver-side cancellations impact operational efficiency

9. Conclusion

- Analyzed OLA ride-booking data to understand overall business performance.
- Used **SQL** to extract insights from raw data by solving real business questions.
- Identified successful, cancelled, and incomplete rides to evaluate operational efficiency.
- Analyzed cancellation reasons to highlight major operational and customer-related issues.
- Studied customer booking patterns to identify top and high-value customers.
- Evaluated vehicle type performance based on ride distance and customer ratings.
- Analyzed customer and driver ratings to measure service quality and satisfaction levels.
- Examined payment methods and found strong adoption of digital payments such as UPI.
- Calculated booking values to understand revenue contribution and trends.
- Built an interactive **Power BI dashboard** to visualize key metrics and insights.
- Dashboard visuals made complex data easy to understand for non-technical users.
- Insights from the project can help improve operational efficiency and reduce cancellations.
- The project supports data-driven decision-making for better business growth.